

Scrutinizing Agile Practices or Shoot-out at the Process Corral!

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ABSTRACT

Agile methods and practices are gaining momentum in industry, and are also slowly making their way in academia bringing fresh air and funny new jargon. Some practitioners consider them as the ultimate advance in software engineering. But what do we know about this? Where is the evidence? Do they scale? Do they solve real issues or just substitute new issues to old ones? Are the benefits tangible, or just acts of faith? Aren't we all agile? Are there no agile failures? Isn't the "waterfall process" - that piñata of agilistas--the real holy grail of software engineering, and agile processes only a Band-Aid to compensate for our deficiencies? This workshop aims at challenging the ready-made ideas, the fluff, the hype, putting things into context, and examining these with fresh and open eyes.

Categories and Subject Descriptors

D.2.9 Management, Software process models (e.g., CMM, ISO, PSP); Productivity

General Terms

Design, Human Factors, Management

Keywords

Agile, software process, software methods, software tools

1. Workshop Overview

During a ski vacation a group of software methodologists signed the agile manifesto dramatically boosting interest in what were then known as "light weight" methods and launched the agile movement. Seven years later, agile methods and practices are gaining momentum in industry, and are slowly making their way in academia bringing with them fresh air and funny new jargon. Some practitioners consider them as the ultimate advance in software engineering and others find them wanting. But what do we really know about this? Where is the evidence? Do they scale? Do they solve real issues or just substitute new issues to old ones? Are the benefits tangible, or just acts of faith?

If agile is to grow beyond what some view as feel good backlash against documentation driven processes then it must be subject to critical examination. This workshop aims at challenging the ready-made ideas, the fluff, the hype, putting things into context, and examining these with fresh and open eyes.

Our critical examination of agile begins with David Gelperin's deconstruction of the agile manifesto and principles. *Exploring Agile* chronicles fictional everyday software developer Bob's attempts to make sense of the agile manifesto and principles. How literally should he interpret these and what are the limits for agile methods?

One impediment to someone like Bob making sense of agile are the numerous misperceptions and myths that seem to swirl around agile; agile is documentation free, there is no design in agile, agility is for free. Hochmüller and Mittermier attempt to defuse and rebut these myths in *Agile Process Myths*.

An unfortunate fallout from the myths, vagueness, and enthusiasm surrounding agile is some choose to define agile by placing it at the opposing end of some continuum to traditional methods creating what has sometimes been referred to as methodology wars. In these methodology wars, an oversimplified caricature of the waterfall often becomes the favourite piñata of some agilistas in their attempt to demonstrate the superiority of their favourite method. In their paper *Towards a Framework for Understanding the Relationships Between Classical Software Engineering and Agile Methodologies*, Jiang and Eberlien attempt to tone down the rhetoric and bring clarity to the debate by proposing a five dimensional framework to examine the concepts and practices of traditional and agile methods. We may discover we are more alike than we are different.

Delivering value is the very core of agile which means delivering those features which are most highly valued by the customer. Most agile methods assume the existence of a near omniscient and empowered customer capable of prioritizing requirements with no assistance other than a regularly

scheduled system demonstration. Somehow we have absolved ourselves of the responsibility of helping the customer make those decisions. In their paper *Using Measurements to Support Real-Option Thinking in Agile Software Development*, Racheva and Daneva propose taking a real option theory perspective to agile software development. Viewing decisions as options helps decision-makers to become aware of alternatives, to compare them and, consequently, to make the decision that insures best fit between the business goals and the current software functionality

Improving communications between stake holders and developers enables a quicker development tempo. In this context, *In Pursuit of Multi-Modal Executable Acceptance Testing Tool* Park and Maurer argue Executable Acceptance Test Driven Development (EATDD) allows for tighter integration between and software requirements and implementation and improves communications between all project stakeholders.

Agile methods have practical origins. As one well known agilest once quipped “someone tried something, it worked, and so they wrote a book about it” No other method exemplifies this statement more than XP. Born as a “Hail Mary” attempt to save a failing project, XP is synonymous with agile. Many are surprised to discover XP is rigorously defined by a set of synergistic practices from which the excitement and wonder of XP emerge. But inflexibility can be the flip side of rigor which can limit XP’s applicability. In *RDP Technique: a Practice to Customize XP* Mirakhorli et. al. propose a method for tailoring XP using a set of rules rather than practices. They demonstrate their approach on a project to link distributed library catalogues.

Failure is the dark family secret in our industry and agile projects are no different. This is unfortunate because understanding what did not work is a more powerful learning experience than understanding what worked. In *Adopting and Agile Methodology: Why it Didn’t Work*, Krasteva and Ilieva present a case study of three teams attempting their first agile project and unlike many case studies, reveal to the community why it didn’t work.

Finally with a manifesto declaring “individual and interactions over process and tools” agile software development research must take a greater interest in behavioral research. In *Video Analysis of Pair Programming*, Hofer provides an example of behavioural research using video analysis, the results of which challenge some strongly held beliefs.

2. Workshop Goals

This workshop puts in the same room the converts, the enthusiasts, the researchers, the skeptics, the proponents of

more “traditional” approaches, those coming from more delicate areas (safety-critical systems, large legacy code), and have an honest debate, behind the hype and the funny lingo. Ultimately the workshop should identify what is established knowledge, what is dubious knowledge, where research could potentially help validate or invalidate some of the agile practices, and some strategies to get there. Converts wanting to play the role of the devil’s advocate are welcome, too. One outcome of the workshop could be a research agenda on agile practices in software engineering.

3. Workshop Format

The accepted papers will be put online before the workshop takes place, and the prospective participants will be invited to read them as preparation for the workshop. This will save time and will allow the workshop not to have long paper presentations. There will be only one presentation session consisting of short position statements by some of the authors whose papers have been selected. This session will provide a number of potential topics to be further discussed. The participants will select the most controversial topics to be discussed in the rest of the workshop. A few working groups (3-5) will then be formed, trying to blend opponents and proponents, where each group will undertake some closer analysis. The working groups will reconvene in a short plenary session for the concluding discussions.

4. Organization

4.1 Chairs

Philippe Kruchten, University of British Columbia

Steve Adolph, WSA Consulting Inc

4.2 Program Committee

- Pekka Abrahamsson, VTT, Finland
- Muhammad Ali Babar, LERO, Ireland
- Robert Biddle, Carleton University, Ottawa, Canada
- Torgeir Dingsøyr, SINTEF, Trondheim, Norway
- Hakan Erdogmus, NRC, Canada
- Li Jiang, University of Adelaide, Australia
- Jonathan Kohl, Calgary
- Frank Maurer, University of Calgary, Canada
- Grigori Melnik, Microsoft, Redmond, USA
- Eugene Nizker, Evident Point, Canada
- James Noble, Victoria University of Wellington, New Zealand
- Linda Rising, Phoenix, AZ
- Laurie Williams, North Carolina State University, USA